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#include<iostream>
using namespace std;

int main(){
int i,j,rows,cols;

cout<<"Enter no of rows and cols respectively:";
cin>>rows>>cols;

int m[rows][cols];
cout<<"Enter elements:"<<endl;
for(i=0;i<rows;i++){
for(j=0;j<cols;j++){
cin>>m[i][j];
}

}

cout<<"Elements are:"<<endl;
for(i=0;i<rows;i++){
for(j=0;j<cols;j++){
cout<<m[i][j]<<"\t";
}
cout<<endl;
}

cout<<"Row major:"<<endl;
for(i=0;i<rows;i++){
for(j=0;j<cols;j++){
cout<<m[i][j]<<"\t";
}
}
cout<<endl;

cout<<"Column major:"<<endl;
for(j=0;j<cols;j++){
for(i=0;i<rows;i++){
cout<<m[i][j]<<"\t";
}
}

cout<<endl;

cout<<"Upper triangular matrix:"<<endl;
for(i=0;i<rows;i++){
for(j=0;j<cols;j++){
if(i<=j)
cout<<m[i][j]<<"\t";
}
}

cout<<endl;

cout<<"lower triangular matrix:"<<endl;
for(i=0;i<rows;i++){
for(j=0;j<cols;j++){

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if(i<j)
cout<<m[i][j]<<"\t";
}
}

cout<<endl;

cout<<"Diagonal matrix:"<<endl;
for(i=0;i<rows;i++){
for(j=0;j<cols;j++){
if(i==j)
cout<<m[i][j]<<"\t";
}
}

cout<<endl;

return 0;
}
```